

Session Objectives

The student will:

- 1. Differentiate normal vs. abnormal findings of the cardiovascular history and physical examination.
- 2. Identify and distinguish abnormal examination findings including murmurs, rubs, and gallops.
- 3. Develop clinical decision-making process by analyzing and incorporating a patient's history and physical examination findings to formulate a thorough differential diagnosis, assessment, and plan.
- 4. Utilize appropriate SOAP note documentation methods relevant to the cardiovascular system.

Cardiovascular Examination

- **Vital signs**
 - HR, BP, BMI, Waist circumference
- **Neck**
 - JVP, Carotid arteries
- **Chest**
 - Cardiac
- **Abdomen**
 - Abdominal aorta
- **Lower extremities**
 - Pulses, edema
- **Special tests**

Vital Signs:

Heart Rate and Blood Pressure

- **HR:** normal = **60-100**
 - Tachycardia **>100**
 - Bradycardia **<60**
- **BP Classification**
→ **JNC 7 has evolved ... let's explore!**

JNC 7 Blood Pressure Classification in Adults Aged ≥ 18 Years			
Category	Systolic		Diastolic
Normal	<120	and	<80
Prehypertension	120-139	or	80-89
Hypertension, Stage 1	140-159	or	90-99
Hypertension, Stage 2	≥160	or	≥100

National Heart, Lung and Blood Institute. *JNC 7 Express. The Seventh Report of the Joint National Committee on the Prevention, Detection, Evaluation and Treatment of High Blood Pressure. 2003*

Neck: Jugular Venous Pressure and Carotid Pulse

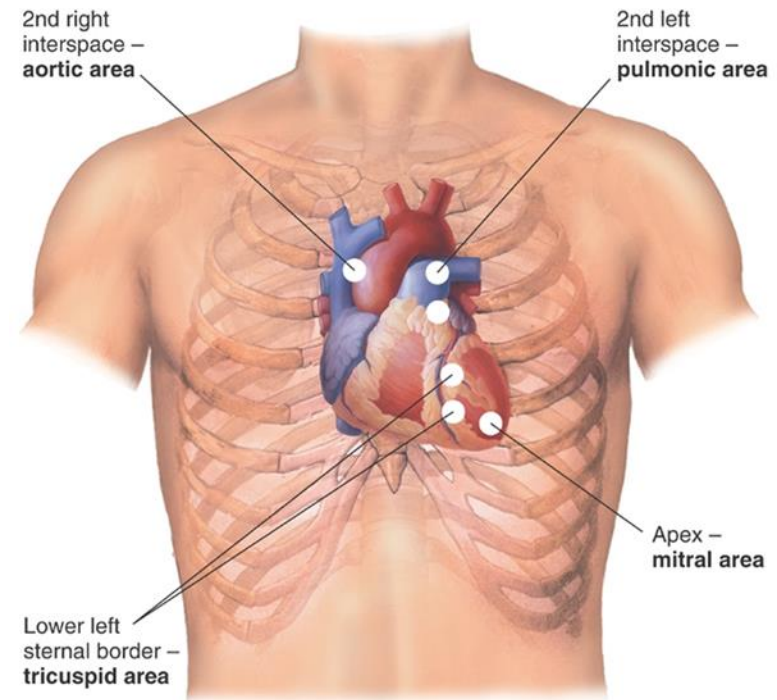
- JVP measured **>3cm (or possibly 4cm) above the sternal angle** → elevated
 - CHF, Pulmonary hypertension, tricuspid stenosis, pericardial compression/tamponade.
- JVP **falls** with hypovolemia and/or loss of blood
- Carotid **upstroke** may be:
 - **Brisk** – normal
 - **Delayed** – suggests aortic stenosis
 - **Bounding** – suggests aortic insufficiency
- Auscultate carotid to assess for **bruits**

The Cardiac Cycle of Systole and Diastole

- **Systole:** the ventricles **contract**
 - The right ventricle pumps blood into the pulmonary arteries (pulmonic valve is **open**)
 - The left ventricle pumps blood into the aorta (aortic valve is **open**)
- **Diastole:** the ventricles **relax**
 - Blood flows from the right atrium → right ventricle (tricuspid valve is **open**)
 - Blood flows from the left atrium → left ventricle (mitral valve is **open**)

Physical Exam - Auscultation

- Aortic Area
 - 2nd ICS, right sternal border
- Pulmonic Area
 - 2nd ICS, left sternal border
- Tricuspid Area
 - 3rd/4th ICS, left sternal border
- Mitral Area
 - 5th ICS, left midclavicular line



The Diaphragm and the Bell

Copyright © 2007 Lippincott Williams & Wilkins. Instructor's Resource CD-ROM to Accompany Bates' Guide to Physical Examination and History Taking, ninth edition.

Cardiac Auscultation

- Auscultate in all listening areas for **S1** and **S2** using the **diaphragm** of the stethoscope
- Then auscultate at the apex with the bell
- The diaphragm and the bell ...
 - The **diaphragm** is best for detecting **high-pitched sounds** like S1, S2, and also S4 and most murmurs
 - The **bell** is best for detecting **low-pitched sounds** like S3 and the rumble of mitral stenosis

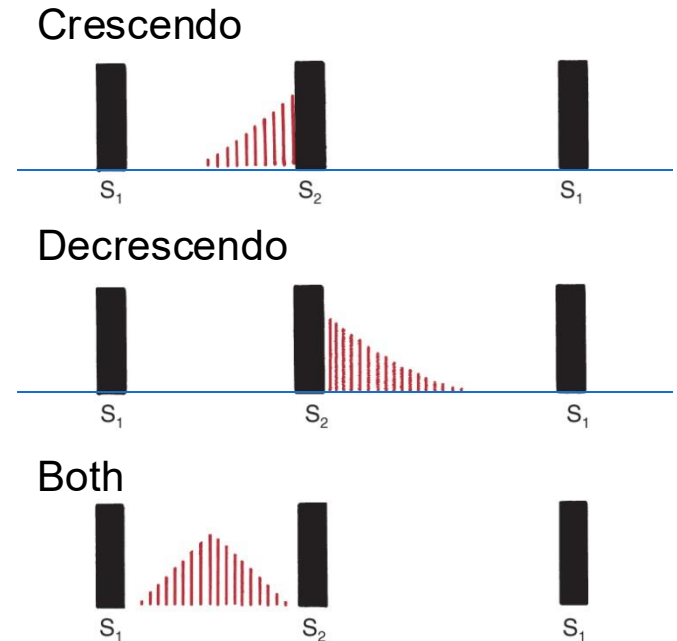


Describing Heart Murmurs: Timing and Duration

- Identify and describe any **murmurs**
- Timing: are the murmurs **systolic** or **diastolic**?
 - **Tip**: palpate the carotid upstroke (occurs in systole) as you auscultate
 - If the murmur coincides with the carotid upstroke, it is **systolic**
- Duration:
 - Early/mid/late systolic (or holosystolic)
 - Early/mid/or late diastolic

Describing Heart Murmurs: Shape

- Shape: Crescendo, decrescendo, or both (“diamond-shaped”)
 - Example: crescendo-decrescendo systolic murmur → **aortic stenosis**
 - Example: Holosystolic murmur → **mitral regurgitation**





Describing Heart Murmurs: Quality, Pitch, and Location

- Quality: Harsh, musical, soft, blowing, or rumbling?
- Pitch: High-, medium-, or low-pitched?
- Examples:
 - Harsh 2/6 medium-pitched holosystolic murmur best heard at the apex → **mitral regurgitation**
 - Soft, blowing 3/6 decrescendo diastolic murmur best heard at the lower left sternal border → **aortic regurgitation**

Assessing the Point of Maximal Impulse

- The PMI may be:
 - Tapping — **normal**
 - Sustained → **LVH [hypertension or aortic stenosis]**
 - Diffuse → **dilated ventricle [CHF or cardiomyopathy]**
- PMI location, amplitude, duration, and diameter:
 - <2.5cm → **normal**
 - Pregnancy → **may displace PMI up and to the left**
 - RVH → **may displace PMI to the right**
 - LVH/Cardiomegaly → **may displace PMI laterally**
 - Pressure overload at LV [aortic stenosis] or volume overload at LV [mitral regurgitation] → **increased amplitude**

Abdomen: Abdominal aorta



- $<3\text{cm}$ → **normal**
- $>3\text{cm}$ → **abdominal aortic aneurysm**

Peripheral Arterial Disease and Ankle-Brachial Index

- ABI → specific and sensitive metric for the diagnosis of PAD.
- ABI has also been shown to predict mortality and adverse cardiovascular events independent of traditional CV risk factors.
- The major cardiovascular societies advise measuring an ABI in every smoker over 50 years old, every diabetic over 50, and all patients over 70.

$$\text{ABI} = \frac{\text{systolic pressure at the ankle}}{\text{systolic pressure at the arm}}$$

ABI Value	Interpretation	Recommendation
Greater than 1.4	Calcification / Vessel Hardening	Refer to vascular specialist
1.0 - 1.4	Normal	None
0.9 - 1.0	Acceptable	
0.8 - 0.9	Some Arterial Disease	Treat risk factors
0.5 - 0.8	Moderate Arterial Disease	Refer to vascular specialist
Less than 0.5	Severe Arterial Disease	Refer to vascular specialist

Clinical Problem Solving: Cardiovascular Complaints



- Always consider life threatening diagnosis
 - Angina
 - Myocardial infarction
 - Dissecting aortic aneurysm
 - Pulmonary embolism
- Etiology *may not be cardiac*
 - GI
 - Pulmonary
 - Musculoskeletal

Cardiac Etiology → Chest Pain

- Left sided chest pain
 - Chest pressure/tightness, sharp pain
- Shortness of breath
- Radiation to the jaw
- Nausea/Vomiting
- Diaphoresis



Atypical Presentation and Symptoms

- Women, geriatric and diabetic patients:
 - Upper back pain
 - Jaw pain
 - Arm pain
 - Shortness of breath
 - Paroxysmal nocturnal dyspnea
 - Nausea/vomiting
 - Fatigue

Understanding the symptoms – Palpitations

- Unpleasant awareness of the heartbeat
- Patients may describe palpitations
 - Skipping
 - Racing
 - Fluttering
 - Pounding
 - Stopping of the heart

- **Cardiac** etiologies
 - Irregular heart beat/arrhythmia
 - Forceful cardiac contraction
- Common *non-cardiac* etiologies
 - Anxiety
 - Hyperthyroidism

- **Dyspnea**: Uncomfortable awareness of breathing that is inappropriate to a given level of exertion
- **Orthopnea**: dyspnea that occurs when the patient is lying down and improves when patient sits up
- **Paroxysmal nocturnal dyspnea**: episodes of sudden dyspnea and orthopnea that awakens the patient from sleep

- Accumulation of excessive fluid in extravascular interstitial space
- Interstitial tissue can absorb liters of fluid
 - “Have you noticed any swelling?”
 - “Where does it occur?”
 - “Is it worse in the morning or at night?”

Assessing Cardiovascular Risks From Patient History

- Past Medical History
 - Dyslipidemia, hypertension, diabetes
- Past Surgical History
 - Previous cardiac surgeries
- Medications/Allergies
 - Compliance with medications
- Family History
 - FH of CVD and age of diagnosis
- Social History
 - Smoking - previous or active
 - Other drug use - previous or active
 - Physical inactivity
 - Unhealthy diet
- Vitals: Weight/BMI/Waist circumference
 - Obesity- especially central

Clinical Case and Chief Complaint

- Setting: Family Medicine Office
- **CHEST PAIN**



Chest Pain Associated Symptoms

- Chest trauma? Denies
- Recent travel? Denies
- Fevers or chills? Denies
- Diaphoresis? **Yes**
- Dyspnea? **Yes**
 - On exertion? **Yes**
 - Orthopnea? No
- Palpitations? No
- Leg swelling? No
- **Nausea/Vomiting? Yes**



Clinical Decision-Making Pearl # 3

- This is your 1st opportunity to target your history questions based on the differential diagnosis you are considering.
- Pertinent negative and positive review of systems will **help to “rule-in or rule-out”** diagnoses in your differential.



Review of Systems

- Constitutional: denies fevers, chills, night sweats, weight loss
- Cardiovascular: (+) *chest pain, diaphoresis, and dyspnea on exertion*; denies palpitations
- Respiratory: (+) *dyspnea on exertion*; denies cough
- Gastrointestinal: (+) *nausea*; denies vomiting, heartburn, abdominal pain
- Musculoskeletal: denies back pain, reproducible chest pain, joint pains
- Psychiatric: (+) *high stress level*; denies anxiety



Clinical Decision-Making Pearl # 4 → Patient Education Opportunity

- Modifiable:

- Former smoker
- Hypertension
- Dyslipidemia
- Obesity
- Poor diet
- Sedentary lifestyle
- Stress

- Non-modifiable:

- Age
- Gender
- Family History



Pertinent Components of History Taking In Patient with Chest Pain

- How does this compare with these activities in the past?
- Are patient's **avoiding certain activities** due to pain or associated symptoms?
- **Detailed social history** (diet, caffeine, etc.)
 - **Ask patient to describe their diet!**
 - What do you typically eat for breakfast, lunch, dinner?
 - Do you feel “uncomfortably stuffed” after eating? (Portion-size)
 - What do you typically snack on?



What are your next steps?

- What is your differential diagnosis for chest pain?
- What is the most likely diagnosis?
- *How did you rule-in or rule-out other diagnoses in your differential?*
 - **In-house testing?**
 - Interpretation of ECG & CXR
 - **History and Physical exam findings?**



Assessment/Plan: Preventative Medicine

- Consider counseling on the following topics:
 - Smoking cessation
 - Dietary modification(s)
 - Physical activity: goal → minimum 150 mins/wk.
- Consider comorbidities:
 - Diabetes, hypertension, and/or dyslipidemia
- Set a follow-up appointment:
 - Reassess risk factors, continue counseling, OMT

Prevention

Counseling

Sample Documentation of Your Physical Exam

O:

- **Vital Signs:**

- **Blood Pressure** (sitting): 110/70 right arm; 108/68 left arm.
- **Radial pulse**: 60 beats per minute, regular.

- **Neck:** No jugular venous distension; No carotid bruits.

- **Heart:** Normal S1, S2; regular rate and rhythm, no murmurs, rubs, or gallops.

- **Extremities:** No cyanosis, clubbing, edema.

O: (*w/ acceptable abbreviations*)

- **Vital Signs:**

- **BP** (sitting): 110/70 (R) arm; 108/68 (L) arm.
- **Radial pulse**: 60 bpm, regular.

- **Neck:** No JVD; No carotid bruits.

- **Heart:** Normal S1, S2; RRR; no M/R/G.

- **Extremities:** No C/C/E.